



Raw Input

Premise: $P = \{p_1, p_2, \dots, p_n\}$
Question: Q

① Hexagonal Semantic Structuring

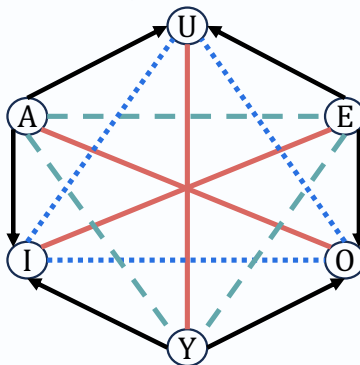


1. Predicate Identification

Extract variable x , subject $M(x)$, property $W(x)$ from Q
 $\mathcal{T}(Q) = \langle x, M, W \rangle$



2. Hexagonal Construction



$\mathcal{H}(Q) = \{A, E, I, O, U, Y\}$



3. Anchor Identification

$\rho(Q) \in \mathcal{H}(Q)$

② Logical Reasoning

Premise: $P = \{p_1, p_2, \dots, p_n\}$
Target: $\rho(Q), \neg\rho(Q)$



1. Translator

NL: $P = \{p_1, p_2, \dots, p_n\} \rightarrow$
FOL: $\Gamma_P = \{\Gamma_{P1}, \Gamma_{P2}, \dots, \Gamma_{Pn}\}$



2. Planner

Goal: $\mathcal{G}_Q = \{G_\rho, G_{\neg\rho}\}$
prove $\rho(Q), \neg\rho(Q)$

Step 1: Identify the Goal

.....

Step n: Conclude the Plan



3. Solver

Step 1: Identify the Goal

.....

Final answer: {True/false/Unknown}

Target: $\rho(Q)$

Verdict: $v(\rho(Q))$
Trace: τ_ρ

Target: $\neg\rho(Q)$

Verdict: $v(\neg\rho(Q))$
Trace: $\tau_{\neg\rho}$

③ Reflective Verification



1. Direct Resolution

$$\rho(Q) = \begin{cases} \text{True,} & \text{if } \rho(Q) = \text{True} \wedge \neg\rho(Q) = \text{False} \\ \text{False,} & \text{if } \rho(Q) = \text{False} \wedge \neg\rho(Q) = \text{True} \\ \text{Unknown,} & \text{if } \rho(Q), \neg\rho(Q) = \text{Unknown} \end{cases}$$


2. Quick Reflection

If $\rho(Q) = \text{True} / \text{False} \wedge \neg\rho(Q) = \text{Unknown}$

$\begin{cases} \text{Type1: } \rho(Q) \text{ correct} \rightarrow \rho(Q) \\ \text{Type2: } \rho(Q) \text{ incorrect} \rightarrow \text{Unknown} \end{cases}$

If $\rho(Q) = \text{Unknown} \wedge \neg\rho(Q) = \text{True} / \text{False}$

$\begin{cases} \text{Type3: } \rho(Q) \text{ correct} \rightarrow \text{Unknown} \\ \text{Type4: } \neg\rho(Q) \text{ correct} \rightarrow \neg(\neg\rho(Q)) \end{cases}$



3. Deep Reflection

If $\rho(Q) \in \mathcal{C}_{ctr} = \{A, E, Y\}$

- $\exists h \in \mathcal{C}_{ctr} \setminus \{\rho(Q)\}, v(h) = \text{True} \rightarrow \text{False}$
- Otherwise $\rightarrow \text{Unknown}$

If $\rho(Q) \in \mathcal{C}_{subctr} = \{U, I, O\}$

- $\exists h \in \mathcal{C}_{subctr} \setminus \rho(Q), v(h) = \text{False} \rightarrow \text{True}$
- $\exists h \in \text{Adj}(\rho(Q)), v(h) = \text{True} \rightarrow \text{True}$
- Otherwise $\rightarrow \text{Unknown}$